Shamsi School
Shamsi Society Campus (Boys' Shift)

Final Examination (2024-25)
Class: VI
Subject: Mathematics

Name: _____Sec: ____Roll #: _____ Invigilator's Sign: ___ T.Marks: 100

OBJECTIVE

Max. Marks: 20

Q1) Fill in the blanks. **/06**

i) The _____ is the distance around a figure.

ii) A set of all integers is an _____ set.

iii) The symbol ϕ represents _____ set.

iv) The number of elements in a set is called its _____.

v) The formula for calculating the area of a rectangle is _____

vi) The perimeter of a rectangle is calculated by the formula _____

Q2) Write one examples of each of the given sets. **/02**

SETS	EXAMPLES
i) Finite Set	
ii) Empty Set	

Q3) Find the square of the following numbers. **/03**

i) 25 = _____

ii) 19 = _____

iii) 21 = _____

Q4) Select the correct option. **/03**

i) Which of these is not a part of a bar graph?

a) Title (b) Y – axis (c) Sector

ii) How many degrees are there in a pie chart?

a) 60 º (b) 90 º (c) 360 º

iii) On the number line which is greater -15 or -3?

a) -3 (b) -15 (c) They are equal

iv) The value of 5 × (- 3) – (- 3) is

a) 18 (b) – 12 (c) – 18

v) If 100% = Rs. 120, then Rs 84 is

a) 120% (b) 70 % (c) 20 %

vi) If 75 % of pupils in a class of 40 passed in a science test, how many of them failed?

a) 10 (b) 25 (c) 30

Q5) State whether the following are true or false. /03

i) A right triangle can also be an isosceles triangle _____.

ii) A scalene triangle is always an acute triangle _____.

iii) An isosceles triangle can also be an obtuse-angles triangle _____.

iv) The exterior angles of an equilateral triangle are not all equal _____.

v) A set of all the children in a school is a finite set _____.

vi) P = { 1, 2, 3, …….., 10 } is written as P = { 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 } in a set – builder form _____.

Q6) Mark each statement in column A with its equivalent in column B. /03

	COLUMN 'A'	**COLUMN 'B'**
i)	A number divided by 6 decreased by 5	(a) 2x + 1
ii)	Twice a number increased by 1	(b) $\frac{x}{6}$ - 5
iii)	8 divided by two-third of a number	(c) 7x + 2
iv)	Seven times a number increased by 2	(d) 6 - $\frac{3}{4}$ x
v)	Three – fourths of a number subtract from 6	(e) 8 ÷ $\frac{2}{3}$ x
vi)	The sum of P and q	(f) P + q

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SUBJECTIVE **Max. Marks:80**

ARITHMETIC SECTION **/36**

Note: Attempt any 6 questions. All questions carry equal marks.

Q1) Simplify:

i) $3\frac{2}{3} \times \frac{3}{11} + [\, 2\frac{3}{4} \div 1\frac{1}{4} \times (-1\frac{2}{3})\,] + (-\frac{1}{3})$

ii) 14 – [16 – { 17 – (18 – 6 – 3) }]

Q2) Find the LCM of the following numbers by prime factorization method.

i) 28 , 35 , 56 , 84

ii) (ii) 35 , 49 , 91

Q3) Calculate the LCM of the following numbers by the method of division.

i) 16, 24, 30, 40, 48

ii) (ii) 14 , 21 , 56

Q4) Convert the following into percentage:

i) $\frac{3}{50}$ (iii) 0 . 35

ii) 4 . 2

Q5) (a) Adila saves 16 % of her monthly salary. If she gets Rs21,000 per month, how much does she save every month?

(b) What percent of 240 is 30?

Q6) Express the first quantity as a percentage of the second quantity.

i) 55 cm , 5 m

ii) 36 min, 6 hrs

Q7) Mr Saleem covered 36 % of a total journey of 150 km while Mr Sabir covered 30% of a total journey of 200 km. Who covered more distance?

Q8) The students of Class V and Class VI are asked about their favorite subject. Information is given in the table below:

Subject	**Class V**	**Class VI**
English	4	7
Urdu	5	5
Mathematics	9	6
Science	7	12

i) What percentage of students of Class V like Mathematics?

ii) What percentage of students of Class VI like Mathematics?

iii) What percentage of students of Class V like Science?

iv) What percentage of students of Class VI like Science?

v) Which subject is the favorite among Class V and Class VI students?

ALGEBRA SECTION /20

Note: Attempt all questions. All questions carry equal marks.

Q9) Simplify:

i) $3\{a - 2(b - \overline{a - b})\}$

ii) $5x - (3x + (4x - 2x))$

Q9) By introducing parentheses evaluate the following.

i) $\frac{5}{2x} + \frac{y}{2x}$

ii) $xy + zx + ax$

iii) $24 - 8y^2 + z^2$

iv) $x^2y + x^2y^2 + x^2y^3$

v) $16 - 2x^2 - 2y^2$

Q10) (a) Twice a number added to half of itself equals 20. Find the number.

(b) Thrice a number decreased by 5 exceeds twice the number by a unit. Find the number.

Q11) Rehan buys a books and a pen and pays Rs 50 for both. The book costs Rs 8 more than twice the cost of the pen. Find the cost of the book.

GEOMETRY SECTION /24

Note: Attempt any 4 questions. All questions carry equal marks.

Q12) Find the perimeter of each of the following:

i) A square of side 8cm

ii) A triangle of side 3cm, 4cm and 5cm

iii) A rectangle of sides 6 cm and 4cm

Q13) Find the area of each of the following rectangular fields whose dimensions are given.

i) Length = 32 m , breadth = 24 m

ii) Length = 4.5 m , breadth = 2m 25 cm

iii) Length = 32 m , breadth = 24 m

Q14) Find the volume of each of the following cuboids.

i) Length = 4 m , breadth = 2.5 m , height = 1.5 m

ii) Length = 12 cm , breadth = 10 cm , height = 8cm

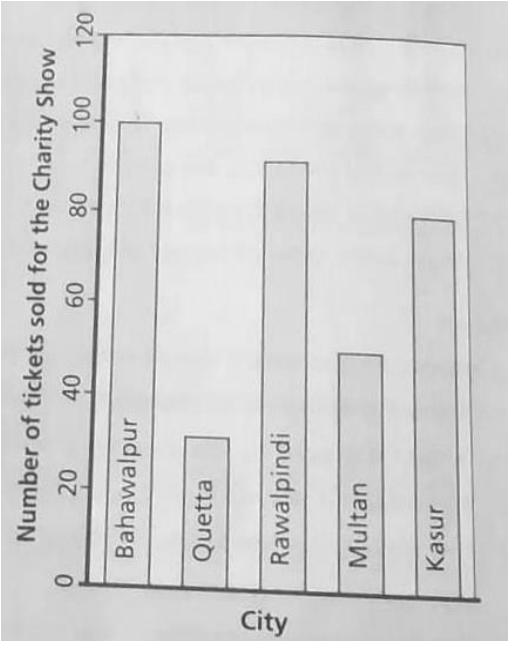
iii) Length = 2 m 50 cm , breadth = 1 m , height = 75 cm

Q15) A tank is 5m long, 3m wide and 1.5m high. How many cubic meters of water can it hold?

Q16) Find the volume of each of the cubes whose edges are given below:

a) 60 cm (b) $2\frac{2}{3}$ m (c) 60 cm

Q17) Read the bar graph and answer the following questions:



i) What is the general information given by the graph?

ii) How many tickets of the charity show was sold by the agents in total?

iii) In which city was the maximum number of tickets sold

iv) In which city was the minimum number of tickets old?

v) The number of tickets sold in Bhagalpur is twice that of tickets sold in Multan. Is this true or false?